RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year: 2021 - 2022 (Even Semester)

INNOVATIVE TEACHING METHODS

Degree, Semester & Branch : VI Semester B.E. ECE B
Course Code & Title : EC8095 VLSI Design
Name of the Faculty member : Mr.A.Rameshbabu

Sl. No.	Date	Topic(s)	Activity*	Reference
UNIT I - INTRODUCTION TO MOS TRANSISTOR				
1.	16.03.22	DC Transfer characteristics	One Minute paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/
				
2.	28.03.22	Stick Diagrams	Write Pair Share	https://www.readingrockets.org/strategies/think-pair-share
				

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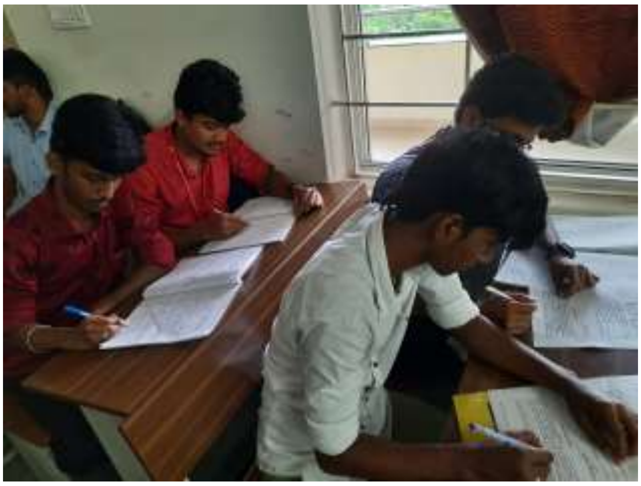
UNIT II - COMBINATIONAL MOS LOGIC CIRCUITS				
1.	08.04.22	Pass Transistor Logic, Transmission Gates	One minute challenge	https://gdhr.wa.gov.au/learning/teaching-strategies/tuning-in/one-minute-challenge
				
2.	18.04.22	Circuit Pitfalls	Mind Map	https://www.ayoa.com/ourblog/6-mind-mapping-examples-for-students-and-teachers/
				

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UNIT III - SEQUENTIAL CIRCUIT DESIGN				
1.	09.05.22	Pipelining, Schmitt Trigger	Head and Talk	https://gdhr.wa.gov.au/learning/teaching-strategies/finding-out/head-talk
				
2.	14.05.22	Timing Issues: Timing Classification of Digital System	One minute challenge	https://icaltefl.com/just-a-minute-speaking-activity/
				

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Sl. No.	Date	Topic(s)	Activity*	Reference
UNIT IV - DESIGN OF ARITHMETIC BUILDING BLOCKS AND SUBSYSTEM				
1.	23.05.22	Multipliers, Shifters	One minute Paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/
				
2.	27.05.22	Designing Memory and Array structures: Memory Architectures and Building Blocks	Open book test	https://www.educationcorner.com/openbook-tests.html
				

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Sl. No.	Date	Topic(s)	Activity*	Reference
UNIT V - IMPLEMENTATION STRATEGIES AND TESTING				
1.	04.06.22	Design for Testability: Scan Design	Head and Talk	https://gdhr.wa.gov.au/learning/teaching-strategies/finding-out/head-talk
				
2.	10.06.22	Boundary Scan & Revision	Mind Map	https://www.ayoa.com/ourblog/6-mind-mapping-examples-for-students-and-teachers/
				

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Innovative Practice Description

- **Unit / Topic:** Unit IV / Case Study: Design as a tradeoff
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 11
- **Activity Chosen:** jigsaw
- **Justification:**
  - ➢ The basic concept of Design as a tradeoff can be explained with Jigsaw method. The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity.
  - ➢ The activity 'JIGSAW' is used to recall the concepts learnt and enhances the learning through this collaboration.
- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**
  - ➢ **Materials for the Activity:**
    The materials for the preparation of the students will be shared one week before through LMS Canvas. Including the material, students were asked to refer to the web content also.
  - ➢ Plan for Implementing this Activity:
    - 5 groups are formed with 5 members
    - Each group is allotted with a name and it's known as Home group.
    - Each expert group is allotted with one problem.
    - The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas.
    - The Expert group members should go to their original shape group and discuss the points (30 Minutes) to the other members.

- All the members in home group learnt about all the topics through expert group members.
- Finally, one from each group should summarize the points of the topic Design as a tradeoff to the class (20 Minutes).
- Review of points and conclusion of the activity (5 minutes)

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 9	PO 10	PO 12	PSO 3
CO3	3	3	3	1	1	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PSO 3
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Jigsaw activity to discuss about power and speed tradeoffs, So Course outcome is mapped at level 1	In Jigsaw activity students will make effective presentations on the design of power and speed tradeoffs,, hence Course outcome is mapped at level 1	Design of power and speed tradeoffs, topic used to Design, analyze and develop electronic products in the area of VLSI Design, So Course outcome is mapped at level 3

- **Images / Screenshot of the practice:**


- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students eagerly participated and enjoyed this activity.
  - ***Benefit of the practice:***
    - Students easily understand the concepts Design of power and speed tradeoffs in VLSI Circuits.
    - Students can answer easily about Design procedures of tradeoffs.
  - ***Challenges faced in implementation:***
    - Lack of preparation: Making the students to learn the materials is the challenging task.
    - Each student has to read the material posted in the course website.
    - Non participation in the activity: All the students should be made involve in the activity
    - Time management: The Jigsaw activity should be able to complete in the planned duration.

References:

- https://computationstructures.org/lectures/tradeoffs/tradeoffs.html
- https://slideplayer.com/slide/1654195/
- https://link.springer.com/chapter/10.1007/0-306-47673-8_3?noAccess=true
- http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.640.3432&rep=rep1&type=pdf

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Innovative Practice Description

- **Unit / Topic:** Unit V / FPGA Building Block Architectures
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 13
- **Activity Chosen:** Flipped Classroom
- **Justification:**
  - ➢ A flipped classroom is a type of blended learning where students are introduced to content at home and practice working through it at College.
- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**
  - ➢ This is the reverse of the more common practice of introducing new content at college, then assigning homework and projects to complete by the students independently at home.
  - ➢ In this blended learning approach, face-to-face interaction is mixed with independent study–usually via technology. In a common Flipped Classroom scenario, students might watch pre-recorded videos at home, then come to school to do the homework armed with questions and at least some background knowledge.
  - ➢ The concept behind the flipped classroom is to rethink when students have access to the resources they need most. If the problem is that students need help doing the work rather than being introduced to the new thinking behind the work, then the solution the flipped classroom takes is to reverse that pattern.
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 4	PO 9	PO 10	PO 12	PSO 3
CO3	3	3	2	1	1	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PSO 3
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Flipped Classroom activity to discuss about FPGA Building Block Architectures, So Course outcome is mapped at level 1	In Flipped Classroom activity students will make effective presentations on FPGA Architectures, hence Course outcome is mapped at level 1	Knowledge of FPGA Building Block Architectures is used to Design, analyze and develop electronic products in the area of VLSI Design, So Course outcome is mapped at level 3

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students eagerly participated and enjoyed this activity.
  - ***Benefit of the practice:***
    - Outcome attainment would have increased due to innovative practice over conventional practice.
    - Students easily understand the concepts of FPGA Building Block Architectures.
    - Students can answer easily about different types of FPGA Building Block Architectures.
  - ***Challenges faced in implementation:***
    - Faced issues while make the students to understand about Flipped Class room Activity rules.

References:

- https://canvas.instructure.com/courses/4390263/files/folder/Notes/Unit%205
- https://www.youtube.com/watch?v=Ctiu5Yi2fZE
- NPTEL: https://www.youtube.com/watch?v=ht7nEjNydDU&t=1713s
- https://towardsdatascience.com/introduction-to-fpga-and-its-architecture-20a62c14421c
- https://www.watelectronics.com/fpga-architecture-applications/